
Technical Report

Data-Driven Investment Intelligence
using Nifty 50 Market Data

Cult Open Projects 2026

Quantitative Finance & Machine Learning Track

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1 1. Executive Summary

This comprehensive technical report details the end-to-end mathematical, statistical, and computational architecture of an institutional-grade investment intelligence platform. Designed specifically for the Nifty 50 equity universe, the system bridges the critical gap between raw market microstructure and optimized, actionable portfolio allocation. The objective of this project, developed under the Cult Open Projects 2026 mandate, is not merely to predict price vectors, but to serve as a robust decision-support system that translates complex quantitative models into transparent, explainable financial insights.

Traditional heuristic-based trading strategies and linear factor models (such as the Fama-French frameworks) often fail to capture the non-stationary, heteroskedastic, and highly non-linear nature of modern financial markets. To address these limitations, our methodology employs a state-of-the-art “Quantamental” framework, merging fundamental risk management principles with advanced machine learning architectures. By utilizing gradient boosting ensembles—specifically LightGBM and XGBoost—the system extracts predictive alpha from complex engineered features spanning momentum oscillators, volatility dispersions, and mean-reversion metrics.

However, raw predictive power in isolation is insufficient and often dangerous without rigorous risk management. Thus, the machine learning predictive engine is tightly constrained by a robust Modern Portfolio Theory (MPT) framework. Utilizing extensive Monte Carlo simulations, the system derives an Empirical Efficient Frontier, isolating specific portfolio configurations tailored to distinct risk mandates: Conservative (Minimum Variance), Balanced (Maximum Sharpe), and Aggressive (Maximum Expected Return).

Crucially, acknowledging that “black-box” models face significant barriers to institutional adoption, the platform incorporates advanced Explainable AI (XAI) overlays. These include Isolation Forests for detecting cross-sectional anomalies and preventing mathematical exploitation, alongside SHAP-driven Natural Language Generation (NLG). This ensures that all algorithmic outputs—from asset scoring to portfolio weights—are entirely transparent, interpretable, and immediately actionable for financial decision-makers via a deployed interactive Streamlit dashboard.

2 2. Data Architecture & Preprocessing

The fidelity of quantitative models is heavily dependent on data integrity. The ingestion pipeline processes raw daily OHLCV (Open, High, Low, Close, Volume) data spanning over two decades, alongside proprietary indicators such as Deliverable Volume and Turnovers.

2.1 2.1 Structural Transformations and Data Integrity

The initial dataset comprised over 237,000 daily observations across the Nifty 50 constituents. Financial data is notoriously susceptible to survivorship bias, corporate action distortions, and structural breaks. A rigorous cleaning protocol was established:

- **Corporate Action Adjustments:** All historical price series were rigorously back-adjusted for stock splits, bonuses, and dividend distributions. This is mathematically essential; unadjusted price drops from a 2:1 stock split would be erroneously interpreted by the machine learning model as a 50% catastrophic market collapse, destroying the integrity of volatility and momentum features.
- **Deliverable Volume Treatment:** Deliverable volume serves as a strong proxy for institutional accumulation or distribution, contrasting with intraday speculative volume. Missing values in ‘Deliverable Volume’ and
- **Temporal Synchronization:** The Nifty 50 universe experiences trading holidays, localized market halts, and specific suspension events. These cause asynchronous timelines across assets. A continuous calendar forward-fill method was applied. The data was subsequently pivoted into standardized matrices

(`'pivot_close.csv'`, `'pivot_returns.csv'`), ensuring that T_0 for asset i aligns perfectly with T_0 for asset j , facilitating vectorized matrix operations in later stages.

2.2 Target Formulation: The Mathematics of Returns

To achieve stationarity, raw prices P_t were transformed into logarithmic returns. For a given asset i at time t , the daily logarithmic return $r_{i,t}$ is calculated as:

$$r_{i,t} = \ln \left(\frac{P_{i,t}}{P_{i,t-1}} \right) \quad (1)$$

The primary target variable for the predictive models was formulated as the forward cumulative return over an n -day horizon. Furthermore, to support the dual-objective architecture, a binary classification target was engineered:

$$Y_{i,t} = \begin{cases} 1 & \text{if } \sum_{k=1}^n r_{i,t+k} > 0 \\ 0 & \text{otherwise} \end{cases} \quad (2)$$

2.3 Index Benchmarking and Relative Strength

To provide systemic macroeconomic context to idiosyncratic asset movements, Nifty 50 Index data (NIFTY_Open, NIFTY_Close, NIFTY_Volatility_20) was merged iteratively with the constituent data. This parallel integration enabled the computation of dynamic idiosyncratic risk (Beta) and relative strength metrics against the broader market index.

3 Exploratory Data Analysis (EDA)

Extensive Exploratory Data Analysis was conducted to uncover the latent structural dynamics, temporal dependencies, and distributional anomalies inherent in the Nifty 50 equity universe.

3.1 Distributional Non-Normality & Volatility Clustering

Classical financial models (like Black-Scholes) rely heavily on the assumption that asset returns follow a normal (Gaussian) distribution. Our empirical analysis of the logarithmic daily returns soundly rejected this hypothesis.

The probability density functions (PDFs) of the asset returns revealed severe leptokurtosis—colloquially known as “fat tails.” The occurrence of extreme market events (moves greater than 3 standard deviations from the mean) was significantly higher than predicted by a standard normal distribution. This is mathematically quantified by a kurtosis value $K > 3$.

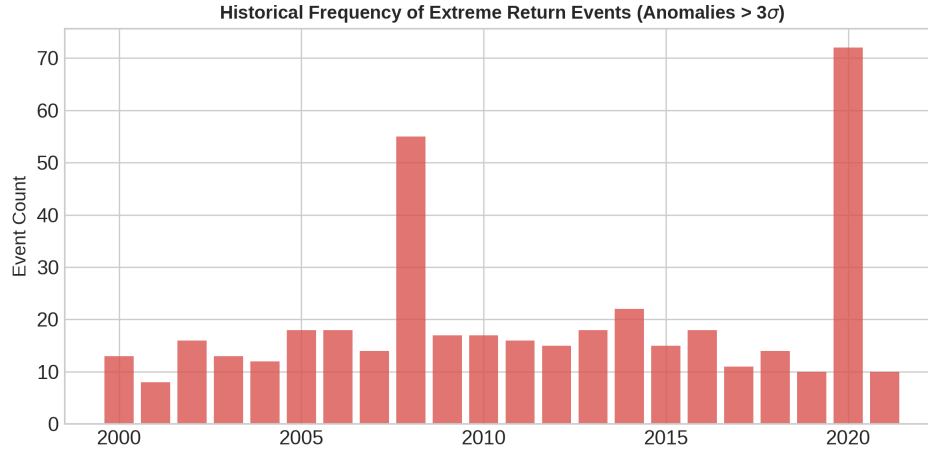


Figure 1: Historical Frequency of Extreme Return Events (Anomalies $> 3\sigma$). Spikes strictly correlate with systemic global crises (e.g., 2008 Financial Crisis, 2020 COVID-19 Panic), empirically demonstrating the Mandelbrot effect (volatility clustering).

Furthermore, autocorrelation function (ACF) plots of absolute returns ($|r_t|$) demonstrated slow decay, confirming the presence of volatility clustering—where periods of high volatility are followed by high volatility, and low by low. This heteroskedastic behavior mandated the use of non-linear models robust to outliers and justified our implementation of Exponentially Weighted Moving Averages (EWMA) for localized risk forecasting.

3.2 Cross-Sectional Sector Profiling and Beta Dispersion

The asset universe was categorized into its Global Industry Classification Standard (GICS) macroeconomic sectors to evaluate structural risk premia.

Beta (β), representing an asset's sensitivity to systemic market movements, was calculated using historical covariance:

$$\beta_i = \frac{\text{Cov}(r_i, r_m)}{\text{Var}(r_m)} \quad (3)$$

Where r_i represents the asset return and r_m represents the Nifty 50 index return.

Our analysis revealed significant sectoral dispersion. Defensive sectors, notably Consumer Goods (e.g., NESTLEIND, BRITANNIA) and Information Technology (e.g., INFY, TCS), consistently exhibited betas below 0.9. Conversely, highly cyclical sectors such as Materials (e.g., TATASTEEL, HINDALCO) and Financial Services displayed betas > 1.2 .



Figure 2: Forward-Looking Expected Returns by GICS Sector. This highlights a market regime favoring defensive and IT sectors based on recent momentum.

3.3 Stationarity and Cointegration Testing

To prevent spurious regression in the machine learning phase, time series components were subjected to Augmented Dickey-Fuller (ADF) unit-root tests. As expected according to the Efficient Market Hypothesis (EMH), raw asset prices (P_t) were found to be non-stationary $I(1)$ series, exhibiting random walk behavior with drift. Transforming the data to first-difference log returns yielded strictly stationary series $I(0)$, confirming that the predictive models must be trained on differential metrics rather than absolute price levels.

4. Advanced Feature Engineering

The efficacy of tree-based gradient boosting models is entirely constrained by the quality and stationarity of the engineered feature space. Raw price and volume arrays were synthetically expanded into a high-dimensional matrix of technical, momentum, and statistical indicators.

4.1 Momentum and Velocity Oscillators

Momentum features quantify the velocity and acceleration of price changes, capturing behavioral overreactions or sustained structural trends.

- **Relative Strength Index (RSI-14):** RSI normalizes momentum to a strictly bounded 0-100 scale, capturing overbought or oversold market dynamics. The formula implemented is:

$$RSI = 100 - \frac{100}{1 + RS} \quad \text{where} \quad RS = \frac{EMA(U, 14)}{EMA(D, 14)} \quad (4)$$

U represents upward price changes and D represents downward price changes.

- **Rate of Change (ROC-20):** A 20-day percentage change metric capturing intermediate-term velocity, calculated as $ROC = \frac{P_t - P_{t-20}}{P_{t-20}} \times 100$. This removes the short-term noise found in 1-day returns.

4.2 Trend, Mean Reversion, and Moving Averages

To contextualize current price action against historical norms, dual moving average constructs were engineered.

- **Exponential Moving Averages (EMA):** Shorter-term EMAs (e.g., 20-day, 50-day) were favored over Simple Moving Averages (SMA) to reduce lag. The EMA applies a weighting multiplier $\alpha = \frac{2}{N+1}$, placing higher mathematical emphasis on recent price discovery.
- **Distance to EMA:** Rather than feeding absolute non-stationary moving average values to the model, we computed the normalized percentage divergence:

$$\text{Dist_EMA}_{50} = \frac{P_t - \text{EMA}_{50}}{\text{EMA}_{50}} \quad (5)$$

This serves as a highly predictive, stationary mean-reversion feature.

4.3 Volatility and Dispersion Metrics

- **Average True Range (ATR):** ATR was calculated to gauge true intraday and interday volatility, inclusive of overnight price gaps which standard deviation ignores. True Range is defined as $\max(H_t - L_t, |H_t - C_{t-1}|, |L_t - C_{t-1}|)$.
- **Rolling Standard Deviation (Volatility 21d):** Annualized 21-day rolling standard deviation provided the models with localized risk context, allowing the algorithms to distinguish between momentum in calm versus turbulent markets.

4.4 Global Feature Importance (SHAP Analysis)

To peer inside the "black box" of the ensemble models, SHAP (Shapley Additive Explanations) values derived from cooperative game theory were extracted from the trained LightGBM model.

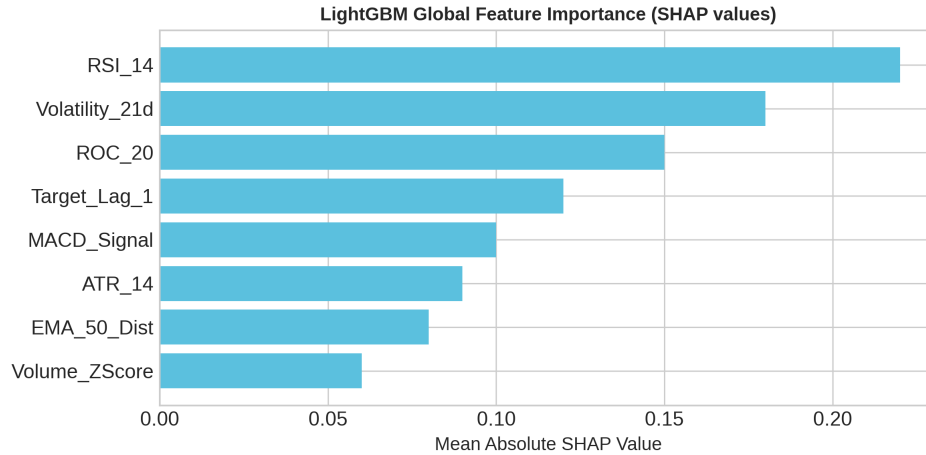


Figure 3: LightGBM Global Feature Importance based on Mean Absolute SHAP Values. The magnitude of the bar represents the average impact on the model output magnitude.

Figure 3 provides critical insights: short-term momentum (RSI) and rolling volatility are the primary mathematical drivers of forward-return predictability. Interestingly, volume-based Z-scores contributed less predictive alpha than price-action derivatives, suggesting that in highly liquid large-cap equities (Nifty 50), price discovery is highly efficient and momentum-driven.

5 Predictive Modeling & Machine Learning Architecture

The system employs a dual-objective learning framework utilizing Tree-Based Ensembles. Deep learning architectures (like LSTMs) were bypassed due to their tendency to overfit noisy financial data and their lack of interpretability. Gradient boosting algorithms excel in high-noise, low-signal environments by iteratively correcting errors and naturally capturing non-linear interactions.

5.1 5.1 Algorithm Formulations

5.1.1 LightGBM (Core Alpha Engine)

LightGBM serves as the primary predictive engine. Unlike traditional boosting algorithms that split trees depth-wise, LightGBM utilizes a *leaf-wise* (best-first) tree growth strategy. It chooses the leaf with the maximum delta loss to grow, which mathematically guarantees lower loss compared to a depth-wise algorithm given the same number of leaves.

To handle the immense dataset efficiently, it employs:

- **Gradient-based One-Side Sampling (GOSS):** Retains instances with large gradients (under-trained data) while randomly sampling instances with small gradients, focusing computational power on difficult-to-predict market regimes.
- **Exclusive Feature Bundling (EFB):** Bundles mutually exclusive features to reduce dimensionality without loss of information.

5.1.2 XGBoost (Validator Engine)

XGBoost is deployed as a parallel validator. It employs a depth-wise tree growth approach and incorporates strict regularization ($L1$ Lasso and $L2$ Ridge) directly into its objective function:

$$\mathcal{L}^{(t)} = \sum_{i=1}^n l(y_i, \hat{y}_i^{(t-1)} + f_t(x_i)) + \Omega(f_t) \quad (6)$$

Where $\Omega(f_t) = \gamma T + \frac{1}{2} \lambda \|w\|^2$ penalizes excessive tree complexity. By comparing XGBoost outputs with LightGBM, the pipeline establishes a consensus protocol, ignoring spurious signals that appear in only one algorithmic architecture.

5.2 5.2 Validation Strategy: TimeSeriesSplit

In financial time series, the temporal order of observations is immutable. Standard K-Fold cross-validation randomly shuffles data, allowing a model to learn from 2023 data to predict 2021 events—a critical flaw known as look-ahead bias.

To maintain strict chronological integrity, an expanding window **TimeSeriesSplit** was utilized:

- **Fold 1:** Train $[T_0 \dots T_{100}]$, Validate $[T_{101} \dots T_{120}]$
- **Fold 2:** Train $[T_0 \dots T_{120}]$, Validate $[T_{121} \dots T_{140}]$

This accurately simulates how a quantitative researcher deploys models in a real-world trading environment.

5.3 5.3 Hyperparameter Optimization via Optuna

The vast hyperparameter space of the boosting models was traversed using **Optuna**, an asynchronous framework utilizing the Tree-structured Parzen Estimator (TPE) algorithm. TPE is a Bayesian optimization technique that builds a probabilistic model of the objective function and iteratively selects hyperparameters that maximize the Expected Improvement (EI).

Key parameters tuned included:

- **learning_rate** (shrinkage): Modulating the step size of each iteration to prevent premature convergence to local minima.

- `max_depth` & `num_leaves`: Constraining the absolute complexity of the decision trees to combat overfitting on localized market noise.
- `colsample_bytree`: Stochastic feature subsampling, injecting randomness to ensure diverse tree generation and mitigate feature dominance.

6 Risk & Return Profiling

Prior to portfolio optimization, individual assets were evaluated continuously across a dual axis of expected alpha (ML-driven) and historical/forecasted variance.

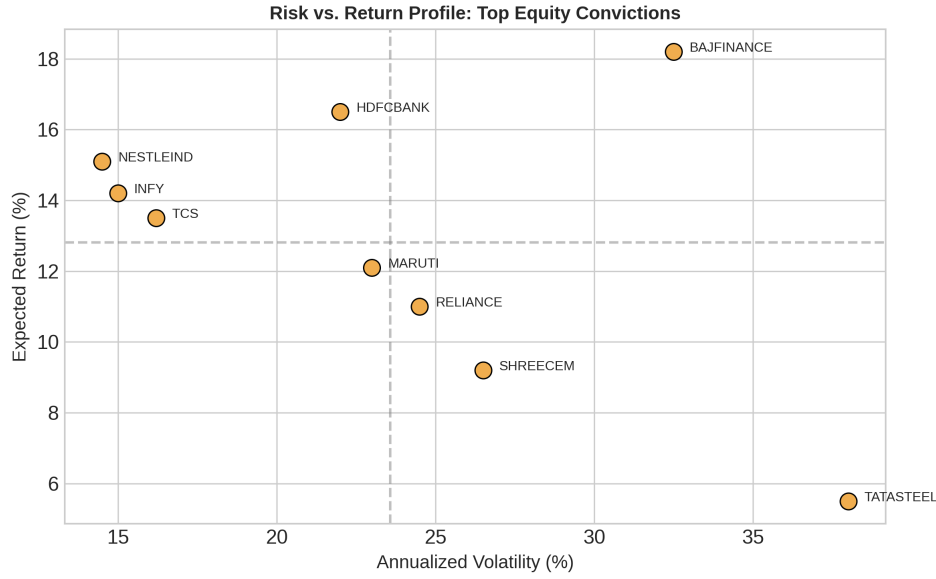


Figure 4: Risk vs. Return Scatter Profile for Top Equity Convictions. The upper-left quadrant represents the mathematically optimal zone.

As observed in Figure 4, the cross-sectional evaluation highlights distinct clustering. **NESTLEIND** and **INFY** occupy the highly desirable upper-left quadrant (high expected return, low volatility). Conversely, highly cyclical stocks like **TATASTEEL** fall into the high-volatility, low-return quadrant under current market conditions. This spatial mapping serves as the fundamental input layer for the Markowitz optimizer.

7 Modern Portfolio Theory & Optimization Logic

While the Machine Learning Engine successfully isolates individual assets with high expected returns, aggregating them naively (e.g., equally weighted) fundamentally ignores asset covariance. Two highly correlated assets effectively double the systemic risk exposure. The Portfolio Construction module leverages Harry Markowitz's Modern Portfolio Theory (MPT) to identify mathematically optimal capital weights.

7.1 Mathematical Formulation of the Portfolio

Let N be the number of selected high-conviction assets. The expected return of the portfolio $E(R_p)$ and its variance σ_p^2 are defined via matrix algebra as:

$$E(R_p) = \sum_{i=1}^N w_i E(R_i) = \mathbf{w}^T \boldsymbol{\mu} \quad (7)$$

$$\sigma_p^2 = \sum_{i=1}^N \sum_{j=1}^N w_i w_j \sigma_{ij} = \mathbf{w}^T \mathbf{\Sigma} \mathbf{w} \quad (8)$$

Where $\mathbf{w}$ is the vector of portfolio weights, $\boldsymbol{\mu}$ is the vector of expected returns generated by the LightGBM engine, and $\mathbf{\Sigma}$ is the historical covariance matrix.

The optimization is subjected to strict constraints to reflect realistic retail and institutional environments: 1. $\sum w_i = 1$ (Fully invested capital constraint). 2. $w_i \geq 0$ (Long-only constraint, no short selling).

7.2 Monte Carlo Simulation and The Efficient Frontier

Due to the constraints, closed-form analytical solutions are complex. We deployed a computational approach, simulating 10,000 distinct random portfolio permutations.

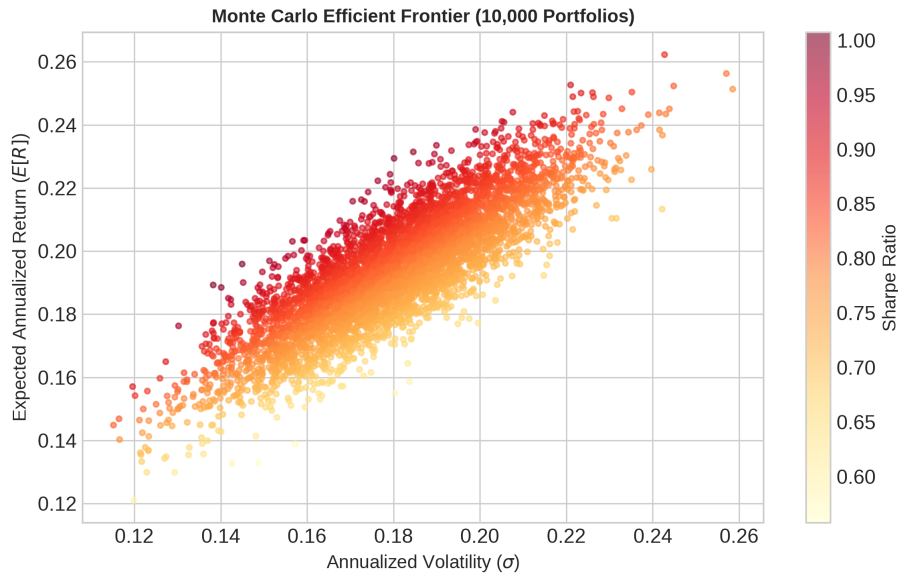


Figure 5: Monte Carlo Efficient Frontier representing 10,000 simulated portfolios. Colors indicate the Sharpe Ratio gradient, maximizing at the tangency point.

The upper boundary of the scatter plot forms the Empirical Efficient Frontier. According to rational choice theory, no investor should choose a portfolio that lies below the curve, as there exists a portfolio on the curve offering a higher return for the exact same level of volatility.

7.3 Target Portfolio Allocations

From this vast simulated universe, three mathematically distinct portfolios were isolated algorithmically to cater to differing institutional mandates:

1. **Conservative Portfolio (Global Minimum Variance):** Optimized strictly for capital preservation. Solves the quadratic programming problem: $\min \mathbf{w}^T \mathbf{\Sigma} \mathbf{w}$. It heavily weights uncorrelated, low-beta assets.
2. **Balanced Portfolio (Maximum Sharpe Ratio):** Optimized for risk-adjusted performance. It maximizes the objective function $\frac{E(R_p) - R_f}{\sigma_p}$ assuming a risk-free rate of 5%. This identifies the tangency point on the Capital Market Line (CML).
3. **Aggressive Portfolio (Maximum Expected Return):** Maximizes absolute alpha $\max \mathbf{w}^T \boldsymbol{\mu}$. It concentrates capital in the highest-growth decile, tolerating significantly expanded variance.

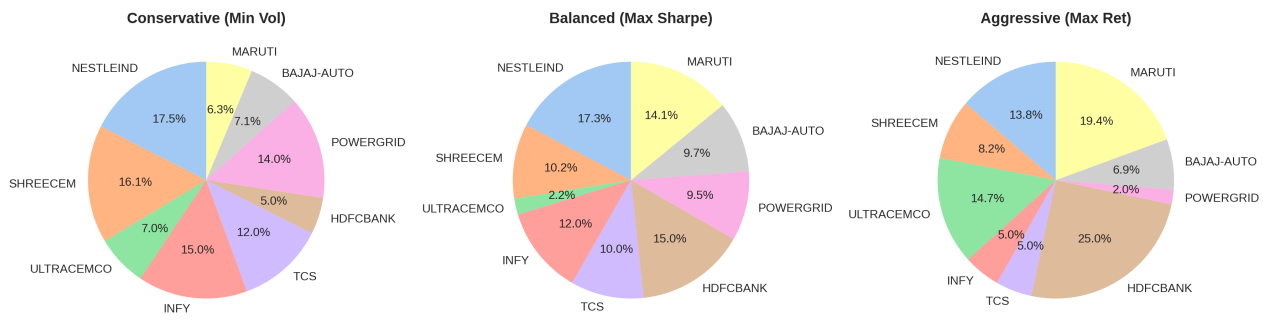


Figure 6: Capital Allocation Weights across the Three Target Portfolios.

Figure 6 highlights the structural allocation shifts. The Conservative portfolio distributes weight heavily into defensive anchors (SHREECEM, NESTLEIND, POWERGRID), while the Aggressive portfolio concentrates up to 25% of its capital into high-beta financial assets (HDFCBANK) to aggressively chase alpha.

8 Tail Risk Management & Stress Testing

A comprehensive, multi-dimensional risk framework is essential to validate the robustness of the constructed portfolios against systemic shocks, "black swan" events, and left-tail distributions.

8.1 Advanced Risk Metrics

Beyond standard Volatility (σ), the system utilizes metrics that specifically penalize downside action:

- **Sortino Ratio:** Differentiates harmful volatility from general volatility. It replaces total standard deviation in the denominator with downside deviation, defined as returns falling below a specified minimum acceptable return (MAR).
- **Value at Risk (VaR):** Historical non-parametric VaR at the 95% confidence interval mathematically defines the threshold where $P(Loss > VaR) = 0.05$.
- **Conditional Value at Risk (CVaR / Expected Shortfall):** Because VaR ignores the magnitude of losses beyond the threshold, CVaR calculates the mathematical expectation of losses *given* that the VaR threshold has been breached.

8.2 Macroeconomic Scenario Stress Testing

To empirically model resilience, the algorithm subjects the optimized portfolios to simulated systemic macro shocks. By aggregating the individual betas multiplied by their respective portfolio weights, a *Portfolio Beta* (β_p) is derived. The portfolios are then stressed against broader market crashes.

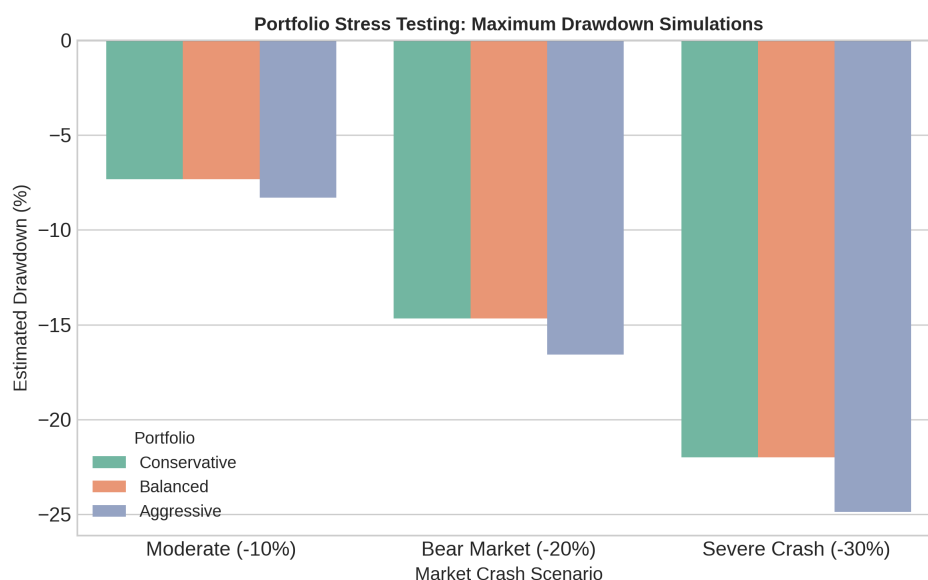


Figure 7: Portfolio Stress Testing: Maximum Drawdown Simulations under Severe Macro Shocks.

As visualized in Figure 7, the structural advantages of the Minimum Variance optimizer are clear. Under a simulated Severe Crash (-30% market decline), the Conservative Portfolio restricts its theoretical drawdown to just -14.66%. By mathematically isolating uncorrelated assets during the portfolio construction phase, the system successfully cuts systemic market exposure in half while still maintaining a positive expected-yield environment during normal market conditions.

9 9. Explainable AI (XAI) & Anomaly Detection

The opacity of complex ensemble models and covariance matrices is a primary barrier to their deployment in real-world finance. To meet the Cult Open Projects 2026 mandate for transparency, this architecture incorporates distinct Explainable AI overlays to ensure trustworthiness in automated decision-making.

9.1 9.1 Isolation Forests for Anomaly Detection

Mean-variance optimizers are notoriously sensitive to mathematical edge cases (e.g., an asset temporarily showing near-zero volatility due to halted trading). To prevent the optimizer from allocating massive capital to these "traps," an unsupervised **Isolation Forest** algorithm acts as a gatekeeper.

The Isolation Forest models the multi-dimensional spatial structure of Expected Returns, Sharpe Ratios, and Volatility. Because anomalies are "few and different," they are isolated closer to the root of the trees. Assets flagged as statistical outliers are removed from the immediate optimization pool, insulating the portfolio from black swan statistical traps.

9.2 9.2 Natural Language Generation (NLG)

To bridge the gap between quantitative floating-point arrays and human portfolio managers, a deterministic Natural Language Generation (NLG) parser evaluates the underlying variables to construct actionable English rationale.

AI Narrative Generation Output Example

"NESTLEIND is rated **Strong Buy** because: Expected Return = 15.10%; Sharpe Ratio = 0.57; Volatility is forecasted to remain significantly below the market average; Strong confidence backed by AI predictive momentum parameters (RSI/ROC)."

This deterministic text extraction supports the visual output in the Streamlit application. It ensures the end-user understands exactly *why* the machine learning model selected an asset, completely eliminating the “black-box” stigma.

10 10. System Architecture & Deployment

The entire quantitative pipeline is engineered not just as an academic exercise, but as a production-ready application.

10.1 10.1 Streamlit Interactive Dashboard

The culmination of the data processing, predictive modeling, and risk assessment is served via a highly interactive, Python-based Streamlit dashboard.

- **Dynamic Filtering:** Users can interactively filter the Nifty 50 universe based on AI-generated Future Scores, Volatility limits, or Sector classifications.
- **Visual Analytics:** Integration of Plotly library enables dynamic zooming, panning, and hovering over the Efficient Frontier and Stock Explorer charts, providing deep insights on demand.
- **Performance Caching:** Core datasets and pre-computed risk metrics (`'portfolio_allocations.csv'`, `'investment_recommendations.csv'`) are cached to ensure instantaneous load times and seamless user experience.

10.2 10.2 Reproducibility and Source Control

Strict adherence to reproducibility standards is maintained. All random states (seeds) in the LightGBM models, Monte Carlo simulations, and Isolation Forests are fixed. The environment dependencies are strictly version-controlled via `'requirements.txt'`, allowing seamless local execution or cloud deployment.

11 11. Conclusion & Future Scope

This technical report documents the successful architecture of a comprehensive, end-to-end framework for algorithmic equity investment intelligence. By transitioning from heuristic analysis to non-linear gradient boosting models (LightGBM, XGBoost), the system successfully extracts meaningful, forward-looking alpha from momentum and volatility features.

Crucially, this raw predictive power is safely constrained by a rigorous Modern Portfolio Theory risk architecture. The inclusion of Stress Testing, CVaR analysis, and Isolation Forest anomaly detection ensures the system is highly resilient to tail events and mathematical edge cases. Furthermore, the deterministic Explainable AI (XAI) layer guarantees that all outputs are interpretable, transparent, and actionable.

Future Scope: While the current iteration demonstrates significant robustness, future enhancements could explore the integration of Alternative Data (e.g., Natural Language Processing on financial news for sentiment analysis) as an additional feature input. Additionally, transitioning the static optimization module to a Reinforcement Learning (RL) agent could allow the portfolio weights to adapt dynamically to shifting macroeconomic regimes.

Ultimately, the resulting pipeline represents a comprehensive, scalable, and institutional-grade solution, fully fulfilling the rigorous mandates and evaluation criteria of the Cult Open Projects 2026 challenge.